News Classification via Textual Analysis and Metadata

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***Abstract*—This paper presents a scalable framework for news classification that combines textual analysis with contextual metadata, such as source reputation and temporal features. By integrating these simple contextual attributes, we develop an approach that enhances document classification accuracy while maintaining high efficiency. Our findings provide a practical roadmap for deploying categorization pipelines in data-heavy editorial environments.**

I. Problem Overview

The primary goal of this project is to develop a machine learning model capable of classifying news articles into seven predefined thematic categories: International News, Business, Technology, Entertainment, Sports, General News, and Health.

The provided dataset consists of approximately 100,000 instances from various international news sources. Each record is characterized by a combination of textual content (title and full article body) and structured metadata, including the publisher, publication timestamp, and the PageRank associated with the source.

The task is structured as a supervised multi-class classification problem where the model must be trained on a development set and evaluated on an unlabeled evaluation set. In accordance with the project requirements, the performance of the pipeline is measured by means of the Macro F1-score, which ensures a balanced evaluation across all news categories.

A. Exploratory Data Analysis

Before defining the methodology adopted to approach the problem, a targeted exploration of the development dataset is essential to identify the informative value of the available features and to uncover potential issues. A significant part of the preliminary analysis was conducted by taking advantage of the VSCode *Data Wrangler* extension, which facilitated the efficient loading of the large development dataset and the extrapolation of rapid yet crucial insights. This exploratory phase, supported by ad-hoc queries for each feature, revealed several key characteristics.

The *source* feature identifies the news publisher for each article. A detailed inspection of the dataset reveals an absence of strictly invalid records, defined as those containing null (*NaN*) or empty values for this feature. However, a significant number of entries contain the string \N as a placeholder. As illustrated in Fig. 1, approximately 70% of the articles in the development set are published by only 10 sources, despite the total number of unique editors exceeding one thousand. This long-tail distribution of the *source* feature suggests that certain publishers have a huge influence on the training data.

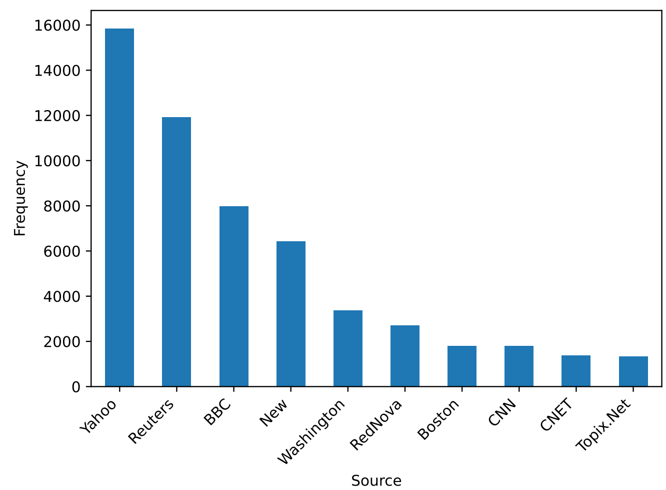

Fig. 1. Top 10 sources distribution

An inspection of the *page_rank* feature, capturing the editor reputation, reveals a high level of data integrity, with no missing values or placeholders identified. However, more than 90% of the entries share a constant PageRank value of five with just four distinct values observed. This lack of statistical variance suggests that the feature may offer limited discriminative power for our classification task.

Proceeding with the analysis of the *timestamp* feature, which identifies the article's publication time, a high level of consistency is once again observed, with no placeholders or invalid values identified. However, a significant anomaly is detected: approximately 35% of the entries contain the value `0000-00-00 00:00:00`. This is interpreted as a temporal placeholder, indicating missing publication data.

The exploratory analysis concludes with an inspection of the *title* and *article* features, which constitute the core textual content of the news. A non-negligible amount of `\N` placeholders was identified within the *article* field. Further manual exploration was conducted using the *Data Wrangler* extension mentioned before to assess text quality. This "byeye" inspection revealed a high density of noise, including HTML entities and residual web artifacts, confirming that while the raw text is highly informative, it requires significant cleaning to be effectively processed by the machine learning pipeline.

II. Proposed approach

A. Preprocessing

During this key phase, the primary objective is to extract meaningful information from each feature while preventing overfitting and capturing the most relevant patterns that define each article category. Regarding the *source* feature, characterized by high cardinality, a significant presence of placeholders, and a long-tail distribution, we employed a `OneHotEncoder` with a frequency-based threshold. Specifically, placeholders and sources with fewer than five occurrences in the development dataset were collapsed into a unified "Other" category. This threshold was strategically chosen to address the trade-off between information retention and feature sparsity: by encoding only 35% of the total unique sources, the model still captures the identity of 98% of the entries with a valid source in the dataset, significantly reducing the dimensionality while preserving the most discriminative signals.

For the *page_rank* feature, the low statistical variance identified in the EDA led to the decision to retain it in its original numerical form. This choice is supported by the nature of the selected classifier (discussed in Section II-B), which effectively manages feature importance through weight optimization during training, allowing it to marginalize the impact of this attribute if redundant.

The *timestamp* metadata was processed to capture cyclical publication patterns and manage the identified placeholder values. We engineered three specific features to represent temporal information:

- *has_date*: A boolean indicator designed to identify records containing the `0000-00-00 00:00:00` placeholder. This feature aims to capture potential correlations between missing temporal metadata and specific editorial sources. For instance, smaller publishers might consistently lack publication timestamps.
- *quarter*: Publication dates were mapped to their respective quarter (ordinal values 1–4). Placeholder records were assigned a value of -1 to maintain numerical consistency while identifying them as a separate group.
- *is_weekend*: A boolean indicator added to capture thematic shifts in news distribution during non-business days, distinguishing weekend patterns from standard weekday publication cycles.

This macro-temporal approach allows the model to capture significant seasonal and weekly trends without introducing the high-dimensional noise and sparsity associated with more granular features, such as the specific hour of publication.

The core of the classification engine focuses on the treatment of the *title* and *article* fields, which constitute the primary information source for category prediction. The process began with a text-cleaning phase, utilizing dedicated functions to remove noise identified during the exploratory analysis. Beyond correcting encoding inconsistencies with `ftfy` and decoding HTML entities, we systematically removed URLs, tags, and non-informative boilerplate sequences such as "read more" or "click here". During this cleaning, the text was converted to lowercase while strategically preserving punctuation, currency symbols, and percentage signs, as these tokens serve as strong statistical indicators for specific categories like *Business*. Notably, explicit stemming or lemmatization steps were deliberately omitted to maintain high computational throughput across the instances and to preserve the semantic integrity of the journalistic register. This choice prevents the potential loss of morphological nuances that can be critical for classification.

Once cleaned, the *title* and *article* fields were merged into a single textual feature to prepare it for the subsequent vectorization process. A heuristic weighting strategy was implemented by concatenating the title twice to the article body, a choice specifically designed to modulate the behavior of the Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer. By doubling the headline, we artificially increase the local term frequency (tf) of its vocabulary. This ensures that the highly discriminative keywords typically found in titles—which provide the most dense signal regarding the article's category—yield a dominant influence on the final numerical representation.

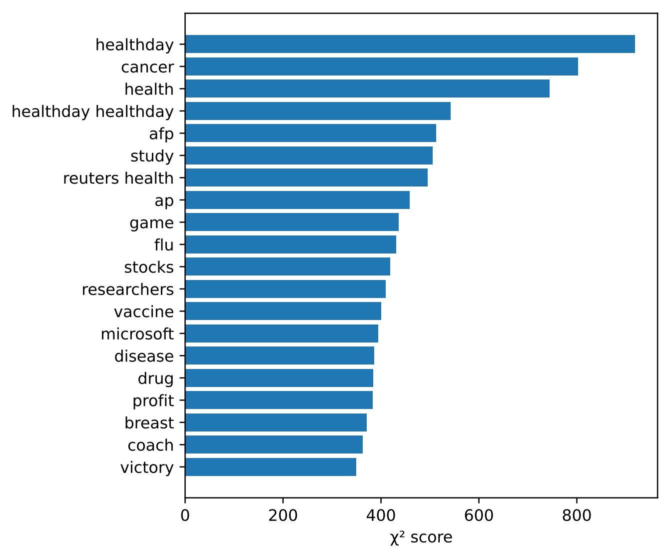

Fig. 2. Top 20 most discriminative TF-IDF terms

Finally, the numerical representation was obtained through a `TfidfVectorizer` configured with sublinear term-frequency scaling and an n-gram range of (1, 2) to capture both individual words and compound phrases. To further refine the feature space, we applied frequency-based filtering: a `min_df` of 3 was set to discard extremely rare tokens that could represent typos or outliers, while a `max_df` of 0.95 was used to exclude generic stop words that appear in almost all documents, thereby carrying negligible discriminative power. Given the high dimensionality arising from a potential space of 400,000 features, a subsequent feature selection step was integrated using the `SelectKBest` algorithm based on the χ^2 (chi-squared) statistic. By reducing the set to the top 80,000 terms, the pipeline not only optimizes computational efficiency but also actively mitigates the risk of overfitting by isolating the linguistic patterns with the highest correlation to the target categories. Figure 2 reports the top 20 most discriminative terms according to the χ^2 score computed on TF–IDF features. Most of the highlighted terms are semantically related to specific news topics, confirming that the classifier primarily exploits content-related cues. Notably, a few highly ranked terms correspond to news sources that were not removed from the textual content; however, their presence does not appear to significantly affect the overall classification performance.

B. Model selection

The choice of the classification algorithm was primarily driven by the characteristics of the resulting feature space, which comprises more than 80,000 features after text vectorization and feature selection. In this setting, linear models are known to offer a favorable trade-off between computational efficiency and generalization ability. We therefore adopted a linear Support Vector Classifier (`LinearSVC`), a decision strongly supported by the literature on text categorization.

As originally demonstrated by Joachims [1], textual data exhibits several properties that make linear SVMs particularly well suited for this task: document representations lie in very high-dimensional spaces, feature vectors are extremely sparse, and most features (i.e., words or n-grams) carry at least some discriminative information. Under these conditions, text classification problems are often close to linearly separable, making the use of complex non-linear kernels or ensemble-based models unnecessary. Indeed, such models (e.g., `RandomForestClassifier` or `XGBoost`) tend to overfit when the number of features greatly exceeds the number of training instances ($p \gg n$), while also incurring a substantially higher computational cost.

Furthermore, compared to regressor-based approaches (e.g., `LinearRegression`, `RandomForestRegressor`), which are conceptually not suited for categorical prediction tasks, the `LinearSVC` explicitly optimizes class separation through margin maximization. Moreover, unlike tree-based learners such as `LightGBM` or `XGBoost`, which struggle to partition a feature space where information is distributed across thousands of weakly informative dimensions, the linear SVM benefits from L_2 regularization, which effectively controls the global distribution of feature weights and mitigates overfitting.

To further account for the pronounced class imbalance present in the dataset, the classifier was configured with `class_weight='balanced'`, assigning each class a weight inversely proportional to its frequency. This choice directly supports the optimization of the Macro F1-score, which was selected as the primary evaluation metric. Finally, the model was trained with `max_iter=10,000` to ensure convergence, and `random_state=42` was set to guarantee reproducibility of the experimental results.

C. Hyperparameters tuning

To identify the optimal configuration for the classification pipeline, we conducted a systematic optimization phase using the `GridSearchCV` framework. This process focused on the two most influential hyperparameters governing the model's structural and statistical complexity:

- Feature Selection Threshold (k): We explored $k \in \{60{,}000, 80{,}000, 100{,}000\}$ within the `SelectKBest` module. This parameter controls the dimensionality of the input space by retaining only the k features with the highest χ^2 (chi-squared) scores. Tuning k is essential to find the optimal balance between providing the model with enough linguistic signals and avoiding the "curse of dimensionality," where excessive noise from irrelevant terms could degrade the performance of the linear classifier.
- Regularization Strength (C): We evaluated $C \in \{0.10, 0.15, 0.25, 0.40, 0.60\}$ for the `LinearSVC`. In Support Vector Machine theory, C acts as a penalty parameter for error terms. It controls the trade-off between achieving a low training error and maximizing the decision margin. A smaller C value encourages a wider margin and a simpler decision function, thereby enhancing the model's robustness and preventing overfitting in high-dimensional sparse spaces.

The optimization process followed a 5-fold cross-validation (CV) strategy to ensure that the performance estimates were robust and not biased by a specific training split. Considering the parameter grid (3 values for k multiplied by 5 values for C), the procedure involved evaluating 15 unique candidate configurations. Given the 5-fold CV, the system executed a total of 75 training iterations.

III. Results

Following the extensive grid search process, the best-performing configuration was obtained with a regularization parameter $C = 0.25$ and a feature selection threshold of $k = 80{,}000$. This setting achieved a Validation Macro F1-score of 0.7003, satisfying the competition requirements and indicating a well-balanced performance across heterogeneous news categories.

Figure 3 reports the confusion matrix computed on the evaluation set and provides an intuitive overview of the model's prediction behavior. The most evident source of error is observed between *International News* (Class 0) and *General News* (Class 5), where a substantial number of instances are mutually misclassified. Specifically, many articles belonging to *General News* are predicted as *International News*, suggesting a significant thematic and lexical overlap between these categories. This behavior is expected, as both classes often cover broadly similar events with limited domain-specific vocabulary.

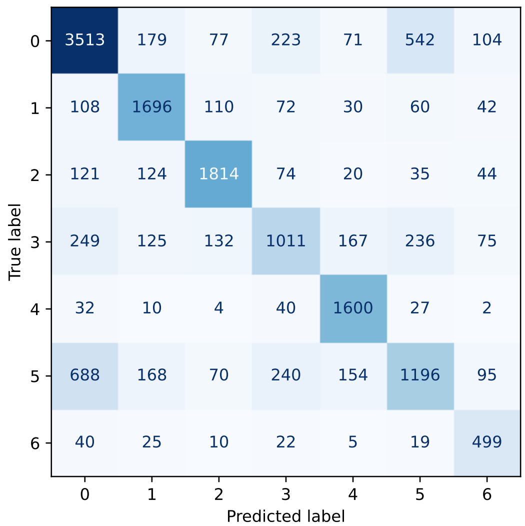

Fig. 3. Confusion Matrix (evaluation set)

In contrast, the model demonstrates strong discriminative capability for more specialized domains. *Technology* (Class 2) and *Sports* (Class 4) exhibit high concentrations along the diagonal of the confusion matrix, confirming that their linguistic patterns are sufficiently distinctive to be captured by a linear decision boundary. Notably, despite being the smallest class, *Health* (Class 6) achieves a high recall, indicating that the adopted `class_weight='balanced'` strategy effectively mitigates class imbalance and prevents the classifier from collapsing towards majority classes.

TABLE I
CLASSIFICATION REPORT ON EVALUATION SET

Class	Precision	Recall	F1-score	Support
0	0.7394	0.7460	0.7427	4,709
1	0.7288	0.8008	0.7631	2,118
2	0.8182	0.8127	0.8155	2,232
3	0.6011	0.5068	0.5499	1,995
4	0.7816	0.9329	0.8506	1,715
5	0.5655	0.4581	0.5061	2,611
6	0.5796	0.8048	0.6739	620
Accuracy		0.7081		16,000
Macro Avg	0.6877	0.7232	**0.7003**	16,000
Weighted Avg	0.7017	0.7081	0.7018	16,000

The quantitative performance is summarized in Table I, which reports precision, recall, and F1-score for each category. The classifier achieves its best results on *Sports* (Class 4), with an F1-score of 0.85 and a recall exceeding 0.93, confirming the presence of highly distinctive terminology. *Technology* (Class 2) also shows consistently strong performance, with balanced precision and recall above 0.81.

Conversely, *Entertainment* (Class 3) and *General News* (Class 5) remain the most challenging categories, with F1-scores of 0.55 and 0.51 respectively. These classes tend to overlap semantically with multiple domains and often rely on more generic language, making them harder to separate using linear TF–IDF representations alone. Nevertheless, the overall Macro F1-score of 0.70 confirms that the model maintains robust performance across both frequent and minority classes.

In order to better contextualize the obtained results, it is worth comparing the achieved performance against the public benchmarks provided for this task. The baseline model released on the public leaderboard reports a Macro F1-score of 0.443, which is substantially outperformed by the proposed approach, yielding an improvement of more than 25 absolute points. This confirms the effectiveness of the adopted feature engineering and classification strategy. At the time of writing, the best-performing submission on the public leaderboard achieves a Macro F1-score of approximately 0.746. While this result remains higher than the score obtained in this work, the gap is relatively limited, especially considering the simplicity, interpretability, and computational efficiency of the proposed linear model. These observations suggest that the current solution represents a strong and competitive baseline, leaving room for further improvements through more advanced representations or modeling techniques.

IV. DISCUSSION

This work demonstrates that a carefully designed linear text classification pipeline, combining TF–IDF representations, χ^2 feature selection, and a `LinearSVC`, can achieve strong and well-balanced performance in a high-dimensional, sparse setting. The results confirm that most of the discriminative power arises from topical lexical cues, while auxiliary metadata features provide complementary signals without dominating the decision process.

Several directions could be explored to further improve performance. First, incorporating contextual representations such as pretrained word embeddings or transformer-based sentence encoders may help resolve ambiguities between semantically overlapping categories like *International News* and *General News*. Second, hierarchical or multi-stage classification strategies could be employed to first separate broad news domains before refining predictions at a finer level. Finally, a more explicit modeling of temporal information or editorial style could provide additional discriminatory power, particularly for challenging classes with diffuse thematic boundaries.

Overall, the proposed approach offers a strong and interpretable baseline for large-scale news classification tasks, balancing effectiveness, computational efficiency, and robustness to class imbalance.

REFERENCES

[1] T. Joachims, "Text categorization with support vector machines: Learning with many relevant features," in *Proceedings of the European Conference on Machine Learning (ECML)*, pp. 137–142, Springer, 1998.